

JourneyAI (Travel Journal) — Complete Project Summary

Simple Language Overview for Project Review

What Is This Project?

JourneyAI started as a basic **Travel Journal** (PHP + MySQL) where users write trip diaries. We upgraded it into a **Smart Travel Planner** that:

- Recommends destinations based on your budget, interests, and travel style
- Predicts how much a trip will cost (with low/high estimates)
- Creates day-by-day itineraries with real places, meals, and stays
- Shows nearby places and builds routes on a map
- Gives you analytics (your spending vs. others, best value destinations)

All of this works offline — no paid APIs, no internet required for the AI parts.

Team & Roll Number Mapping

Roll No	Name	Role (AI/ML)	Role (Web/App)
Roll 1	Member 1	**Data & Corpus** — Database, synthetic data (1,500 journals), destination tags	**Catalog Pages** — View, Edit, Delete, List entries
Roll 2	Member 2	**NLP & Features** — Text processing, sentiment, activity extraction, destination embeddings	**Design System & Landing** — Animations, theme, login/register pages, Project Manager
Roll 3	Member 3	**Recommender** — Hybrid recommendation engine, explanations, "nearby" suggestions	**Trip Planner Page** — Budget slider, interest chips, recommendation cards
Roll 4	Member 4	**Cost & Serving** — Cost prediction model, Flask API, itinerary generator	**Analytics Dashboard** — Charts, personal insights, security, PHP↔Python integration, I

> **Rule:** Every member builds **both** the AI model AND the web page that shows it.

Tools & Technologies (Simple List)

Web Layer (What Users See)

Tool	Purpose
PHP 8.0	Server-side language for all web pages
Apache (XAMPP)	Local web server
MySQL / MariaDB	Database (users, journals, expenses, ratings)
HTML5 + CSS3	Page structure & styling
Vanilla JavaScript + jQuery	Interactive elements
CSS Custom Properties	Design tokens (colors, spacing, fonts)
Lenis + GSAP + ScrollTrigger	Smooth scroll & animations
Chart.js / Chartist	Charts on analytics page
Leaflet.js	Interactive maps on route page

AI/ML Layer (The Brain)

Tool	Purpose
Python 3.10	Language for all ML code
Flask	Lightweight API server (runs on 'localhost:5000')
scikit-learn	ML models (Gradient Boosting, TF-IDF, Cosine Similarity)
pandas + numpy	Data processing
NLTK	Text tokenization & lemmatization
pymysql	Python → MySQL connection
OpenStreetMap (Nominatim + Overpass)	Free geocoding & place lookup
pickle	Save/load trained models

Development & Collaboration

Tool	Purpose
Git + GitHub	Version control
VS Code	Code editor
Postman / curl	API testing
Trello / Notion	Task board
Figma	UI design

Features Implemented — Categorized by Roll Number

Roll 1 — Data & Corpus + Journal Pages

Feature	What It Does	Page(s) Integrated
Database Schema & Migration	Created proper tables: 'journals' view, 'destinations', 'interactions', 'journal_features'	Messages (backend)
Synthetic Data Generator	Auto-creates 1,500 realistic trip journals with costs, photos, ratings — so AI has data to learn from	Powers all AI features
Destination Catalog (93 places)	Curated list with real attractions, INR daily costs, peak seasons, popularity tiers	Plan-a-Trip, Destination Recommendations
Journal CRUD Pages	Create, Read, Update, Delete trip entries with prepared statements (secure)	'new-entry.php', 'my-entries.php', 'display.php', 'update.php', 'delete.php'
Expense Breakdown	Split costs into Food, Transport, Accommodation, Shopping, Fees — stored across journal entries	Journal entry form, Analytics

Gujarat Destinations in Catalog: Ahmedabad, Vadodara, Dwarka, Somnath, Kutch, Diu, Palitana, Statue of Unity, Gir

Roll 2 — NLP & Features + Design System

Feature	What It Does	Page(s) Integrated
Activity Gazetteer	100+ keywords → 21 canonical activities (beach, trekking, food, temples, wildlife, etc.)	Plan-a-Trip interest chips
Sentiment Lexicon	250+ positive/negative travel words + negators/intensifiers — computes sentiment score per journal	Journal features, Recommender quality signal

Destination Normalization	Converts messy user input ("Bombay", "Benares", typos) → canonical names ("Mumbai, India", "Varanasi, India")	Plan-a-Trip
TF-IDF Feature Extraction	Turns journal text into numerical vectors for similarity matching	Recommender (semantic signal)
Per-Destination Profiles	Aggregates: average sentiment, activity histogram, cost priors, peak seasons	Recommender, Cost Model
Design System (journeyai.css)	Ocean/teal theme, glassmorphism cards, smooth scroll, reveal animations, dark/light mode	**Every single page
Landing Page (index.php)	Cinematic hero carousel, animated stats, destination showcase	`index.php`
Auth Pages Re-skin	Modern split-screen login/register with validation	`login.php`, `register.php`, `change-pswd.php`

Roll 3 — Recommender + Plan-a-Trip Page

Feature	What It Does	Page(s) Integrated
Content-Based Scoring	Matches your selected interests (beach, food, history...) against destination activity profiles	Plan-a-Trip results
Collaborative Filtering	"Users like you liked these places" — from rating interactions table	Plan-a-Trip, Recommendations page
Semantic Similarity	Compares your interest text against destination journal narratives (TF-IDF cosine)	Plan-a-Trip results
Hybrid Blending	Weighted mix: 38% Content + 22% Semantic + 30% Collaborative + budget fit + hidden gem boost	Plan-a-Trip, Recommendations
Budget-Aware Filtering	Predicts cost for ALL 93 destinations in one go (~0.05s), filters/stretches by your budget	Plan-a-Trip form submission
Hidden Gem Discovery	Boosts lesser-known destinations that score well (Kutch, Palitana, Gir, Spiti, etc.)	Plan-a-Trip cards show "Hidden Gem" ribbon
Proximity / Near Me	Browser geolocation → region → shows destinations near you	Plan-a-Trip "Use My Location"
Explainable AI	Every card shows: "Why this?" modal with factor scores + sample journal quotes	Plan-a-Trip cards (`ja-cards.php`)
Plan-a-Trip Page	Animated form: budget slider, interest chips, trip mode, origin → live recommendation cards	`plan-trip.php`
Recommendations Page	Personalized picks based on your session history	`recommendations.php`

Roll 4 — Cost Model + Serving + Analytics + Itinerary + Maps

Feature	What It Does	Page(s) Integrated
Cost Prediction Model	Gradient Boosting Regressor predicts **daily cost** → multiplies by days. $R^2 = **0.75**$, $MAE \approx **₹7,900**$	Plan-a-Trip cards, Itinerary page, Cost API
Quantile Bands	10th/90th percentile models give **Low / High estimate** (confidence interval)	Everywhere cost shows
Category Breakdown	Splits total into Food, Transport, Accommodation, Shopping, Fees — per destination ratios	Donut charts on cards & itinerary
Party Size Scaling	Cost scales with `party_size^0.85` (accommodation shares sub-linearly)	Plan-a-Trip, Itinerary
Flask ML Microservice	8 REST endpoints on `127.0.0.1:5000` (health, recommend, predict-cost, itinerary, nearby, route, analytics)	All PHP pages via `ml_client.php`
Day-by-Day Itinerary Generator	Creates Morning/Midday/Afternoon/Evening timeline with real attractions, meals, stays, per-day cost — **r	`itinerary.php`
Dynamic Destination Support	Type ANY place (even not in catalog) → fetches real OSM places → builds itinerary on the fly	Itinerary, Nearby, Route
Nearby Places (OSM)	Nominatim geocode + Overpass API → real POIs around any location, filtered by trip mode (day/weekend)	`plan-trip.php` (Nearby tab), `itinerary.php`
Multi-Stop Route Builder	Nearest-neighbour ordering of places → total km + Leaflet map	`route.php`
Analytics Dashboard	Personal stats (your spend vs community), Practical insights (best value, cheapest months, cost ranges), (	`analytics.php`
Security Hardening	bcrypt passwords (hash-on-next-login), prepared statements everywhere, XSS-safe output, Flask localhost	**All pages**
PHP↔Flask Integration	`ml_client.php` — cURL helper with offline banner, timeout, error handling	Every page calling ML

Page-by-Page Feature Map

Page	Key Features	Roll Owner
`index.php`	Hero carousel, animated stats, destination showcase, theme toggle	Roll 2
`login.php` / `register.php` / `change-pswd.php`	Secure auth, bcrypt, hash-on-next-login, modern UI	Roll 2
`dashboard.php`	Quick actions, recommended-for-you, recent entries, saved plans	Roll 1 + 3
`plan-trip.php`	**Centerpiece** — Budget slider, interest chips, trip mode, origin → recommendation cards WITH cost, explanation	Roll 3 (UI) + Roll 4 (API)
`recommendations.php`	Session-history personalized picks, same card system	Roll 3
`itinerary.php`	Day-by-day timeline, cost donut, highlights, attractions, rebuild button, dynamic OSM fallback	Roll 4
`route.php`	Multi-stop route, Leaflet map, total distance, Google Maps links	Roll 4
`analytics.php`	Personal insights, practical insights (best value, cheapest months), corpus charts (Chart.js)	Roll 4
`new-entry.php`	4-step wizard: Details → Story → Budget → Media/Hashtags (photo/video upload)	Roll 1
`my-entries.php` / `display.php` / `update.php`	Full journal CRUD with prepared statements	Roll 1
`profile.php`	Account management, password change	Roll 2

Database Tables (Simple)

Table	Purpose
`signup`	Users (email, bcrypt password, name)
`db`	Journal core: title, city, country, dates, description
`db1`, `db2`, `db3`	Expense breakdown (food, transport, stay, shopping, fees)
`journals` (VIEW)	Joined view with `true_total`, `duration_days`, category totals
`destinations`	93 canonical destinations: tier, region, base cost, peak months
`interactions`	User ↔ destination ratings/saves (for collaborative filtering)
`journal_features`	NLP output per journal: destination, budget bucket, activities, style, sentiment
`media`, `hashtags`, `media_hashtags`	Photos/videos + hashtags on entries (social layer)

Flask API Endpoints (What the Python Service Does)

Endpoint	Method	Input	Output
`/health`	GET	—	Model status, destination count, cost metrics
`/recommend`	POST	budget, duration, interests, style, month, party, origin	Ranked destinations + cost + explanations
`/predict-cost`	POST	destination, duration, style, month, party	Point cost + low/high + category breakdown
`/itinerary`	POST	destination, days, style, month, party	Full day-by-day plan with timeline, meals, stays
`/nearby`	POST	place, mode, interests	Real OSM places around location
`/route`	POST	place, destination (optional), interests, stops	Ordered stops + total km + map data
`/analytics/summary`	GET	—	Corpus stats, best value, cheapest months, cost ranges

‘/analytics/personal’	GET	eml	Your trips, spend by category, vs community avg
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Gujarat-Specific Highlights

Destination	Type	Best Season	Key Attractions	Est. 5-Day Mid-Range Cost
Ahmedabad	Heritage/City	Oct–Mar	Sabarmati Ashram, Adalaj Stepwell, Jama Masjid	~₹10,000
Kutch	Desert/Culture	Nov–Feb (Rann Utsav)	White Rann, Kalo Dungar, Handicraft Villages	~₹12,000
Dwarka	Spiritual	Oct–Mar	Dwarkadhish Temple, Bet Dwarka, Nageshwar Jyotirlinga	~₹9,000
Somnath	Spiritual/Coastal	Oct–Mar	Somnath Temple, Triveni Sangam, Beach	~₹9,500
Gir	Wildlife	Dec–Mar	Asiatic Lion Safari, Devaliya Park	~₹15,000
Diu	Beach/Relax	Oct–Mar	Nagoa Beach, Diu Fort, Naida Caves	~₹9,500
Palitana	Spiritual/Trekking	Oct–Mar	Shatrunjaya Hill (863 temples), Jain Temples	~₹7,500
Statue of Unity	Monument/Nature	Oct–Mar	Statue, Sardar Sarovar Dam, Valley of Flowers	~₹13,000

How to Run (One-Command Setup)

1. Start XAMPP (Apache + MySQL)
2. Database
mysql -u root project < sql/migrations.sql
mysql -u root project < sql/media.sql
3. Python Environment (once)
py -3.10 -m venv ml/venv
ml/venv/Scripts/python.exe -m pip install -r ml/requirements.txt
4. Build Everything (migrate → seed → extract → train → serve)
ml/venv/Scripts/python.exe -m ml.rebuild_all --seed 42 --n 1500
5. Start ML Service
ml/venv/Scripts/python.exe -m ml.app
OR double-click: ml/start_service.bat
6. Open App
http://localhost/travel_journal/index.php

Key Metrics for Review

Metric	Value
Destinations in Catalog	93 (8 Gujarat)
Synthetic Journals	1,500
Users in Corpus	~40
Rating Interactions	~4,000
**Cost Model R ² **	0.75
Cost Model MAE	₹7,900
Recommendation Latency	~300ms (batched)
Pages in Unified Theme	15+
Flask Endpoints	8
Security	bcrypt + prepared statements + XSS-safe

What to Demo Tomorrow

1. **Landing Page** — Cinematic carousel, smooth scroll
2. **Plan-a-Trip** — Set budget ₹15,000, 5 days, interests: beach + food → see Gujarat picks (Diu, Kutch) with cost bands, "Hidden Gem" ribbon, click "Why this?"
3. **Itinerary** — Pick "Kutch, India", 5 days → see day-by-day plan with White Rann, Kalo Dungar, meals, stays, cost donut
4. **Route** — Type "Bhuj" → see multi-stop route on map
5. **Analytics** — Personal spend vs community, best value destinations
6. **Journal Entry** — Create new trip with photo upload + hashtags

All set! This document covers tools, tech, features, roll-wise ownership, page integration, and Gujarat specifics — in simple language.